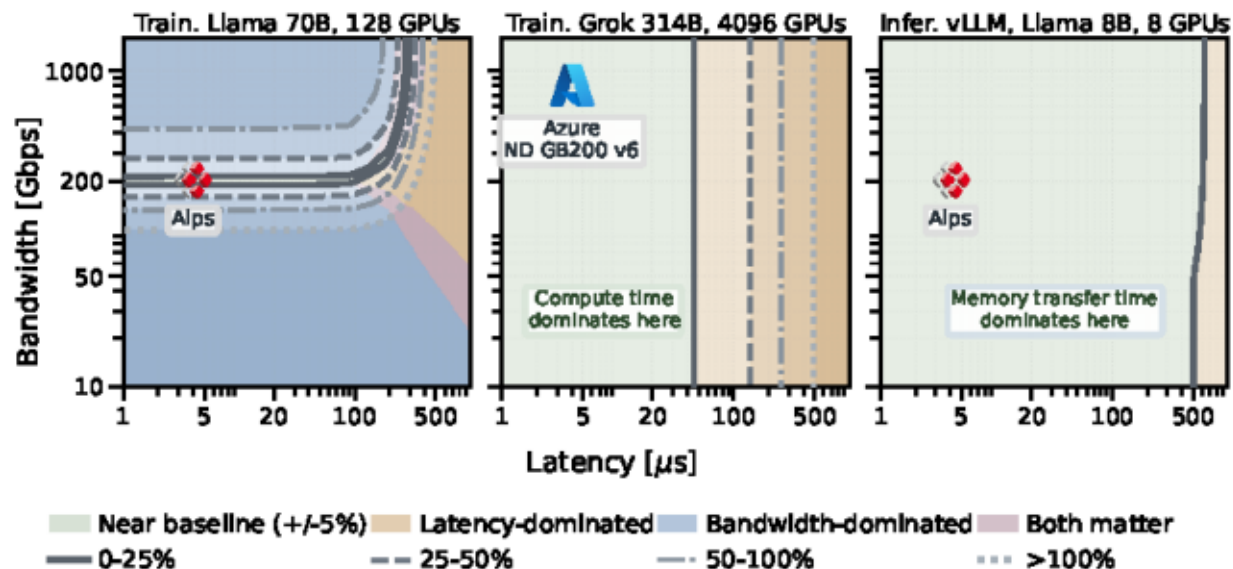


Figure 1: workload sensitivity maps

Paper plot



Artifact output: PLOT-ONLY from committed inputs

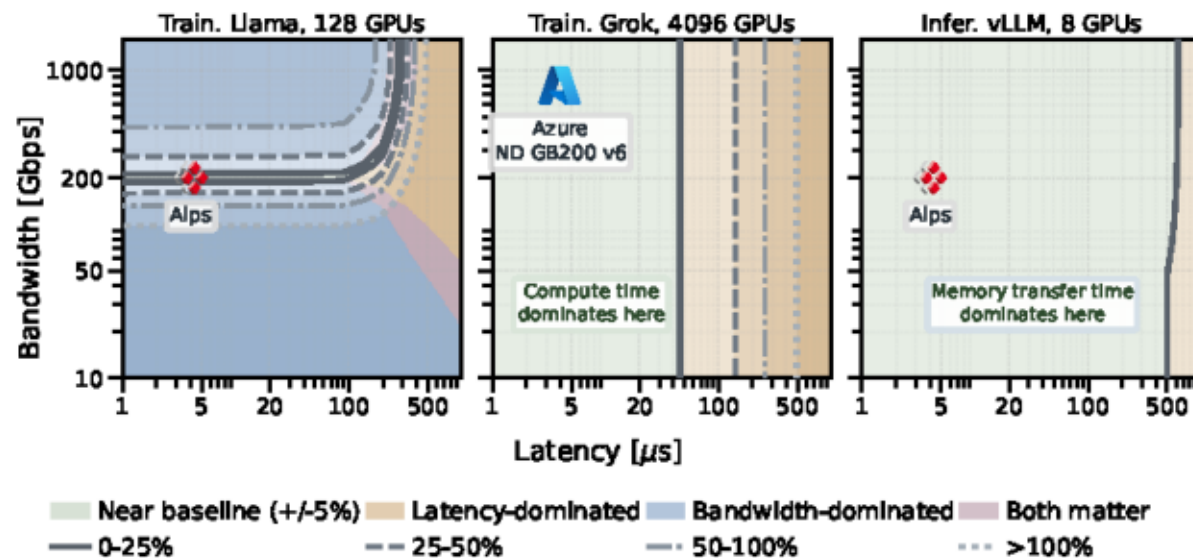
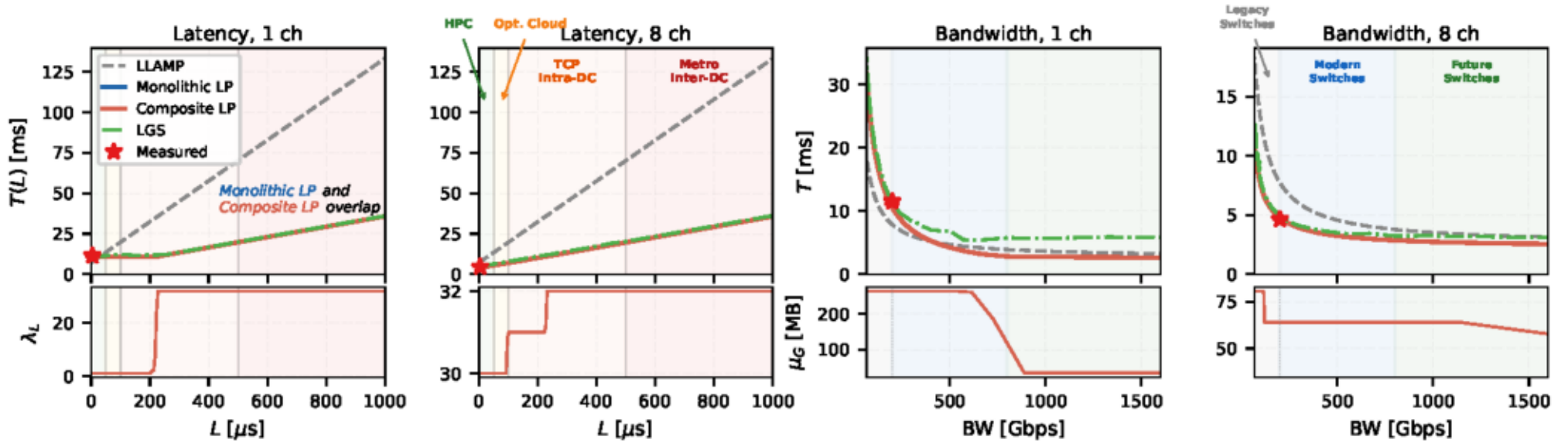
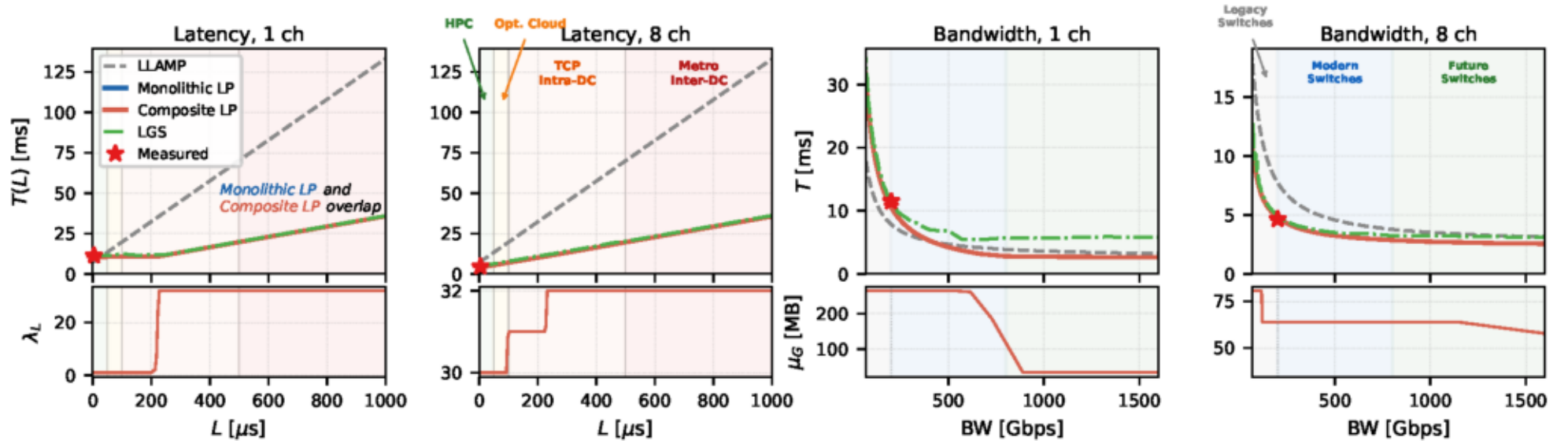


Figure 3: 128 MiB AllReduce

Paper plot



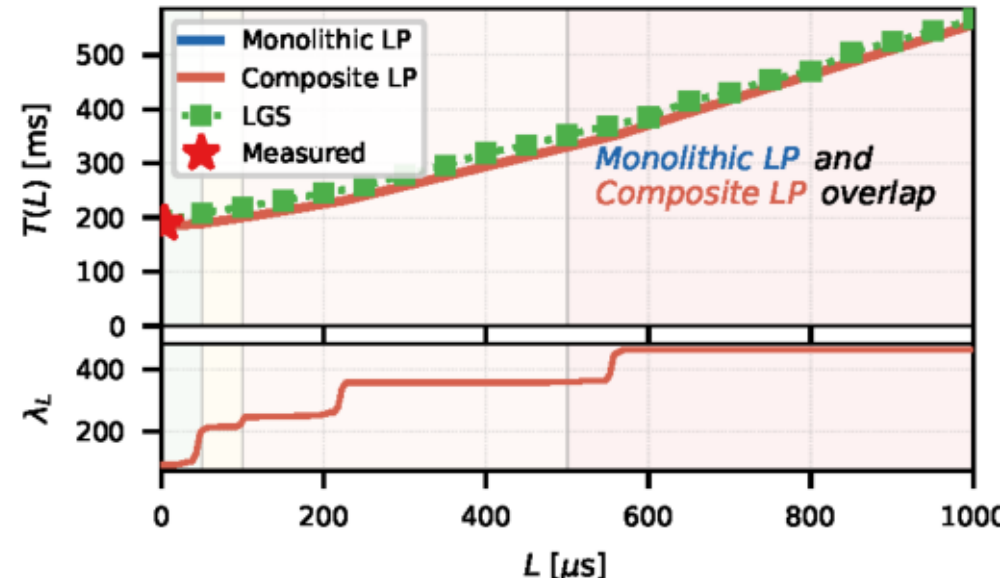
Artifact output: PLOT-ONLY; raw NSYS validation is separate



PROVENANCE: PLOT-ONLY for the displayed paper-style panel. Separate validation CSVs were produced by an NSYS -> SQLite -> GOAL -> LP workflow; building this PDF does not rerun that workflow. Maximum relative errors: 0.078% and 0.166% for latency; 2.373% and 1.493% for bandwidth.

Figure 4: mixed collectives

Paper plot



Artifact output: PLOT-ONLY; raw NSYS validation is separate

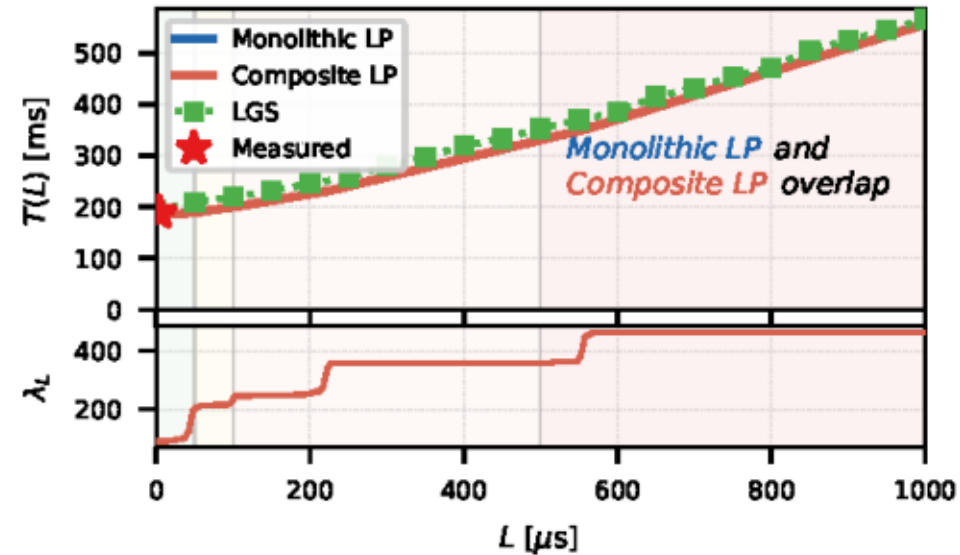
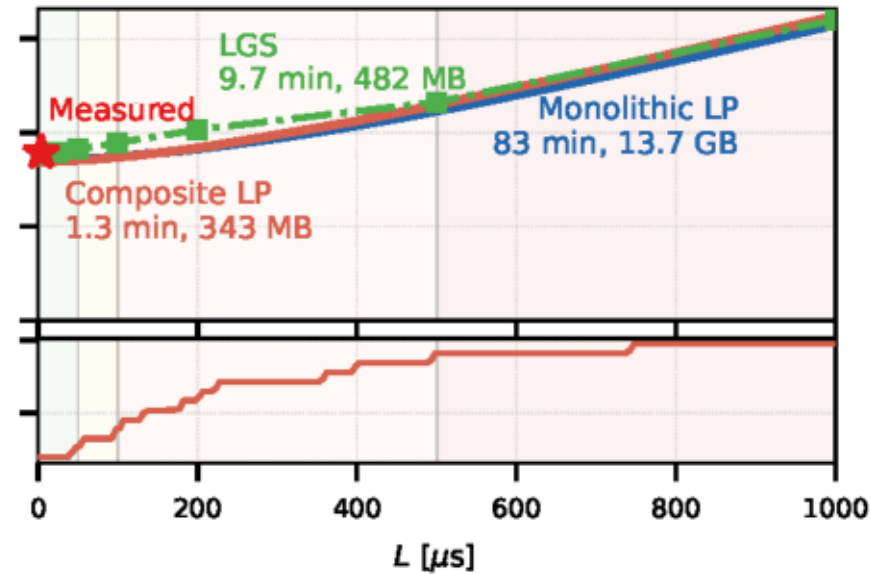
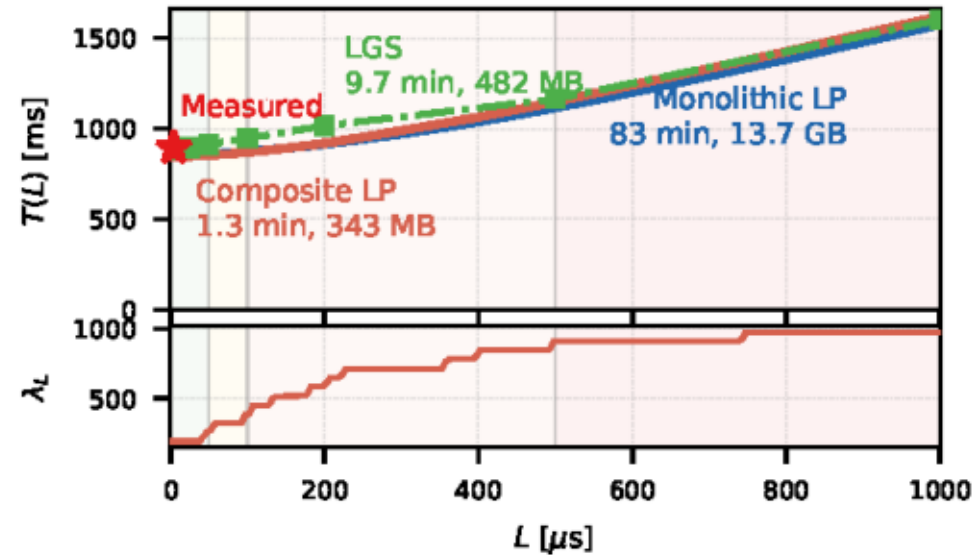


Figure 5: Llama training iteration

Paper plot



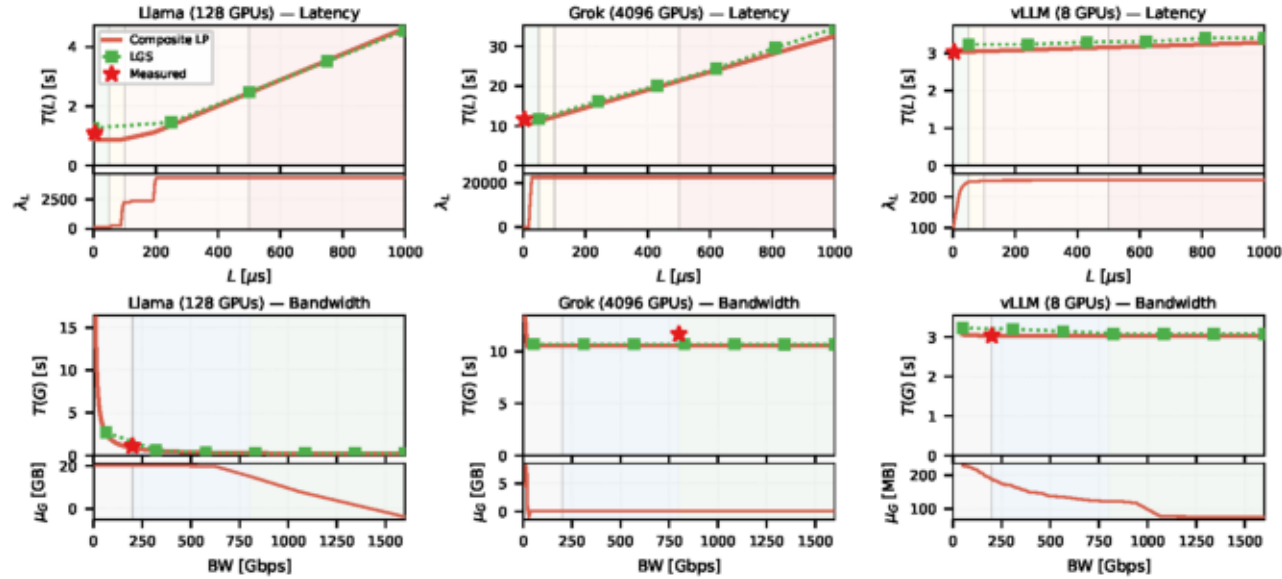
Artifact output: selected NSYS-derived Composite curve



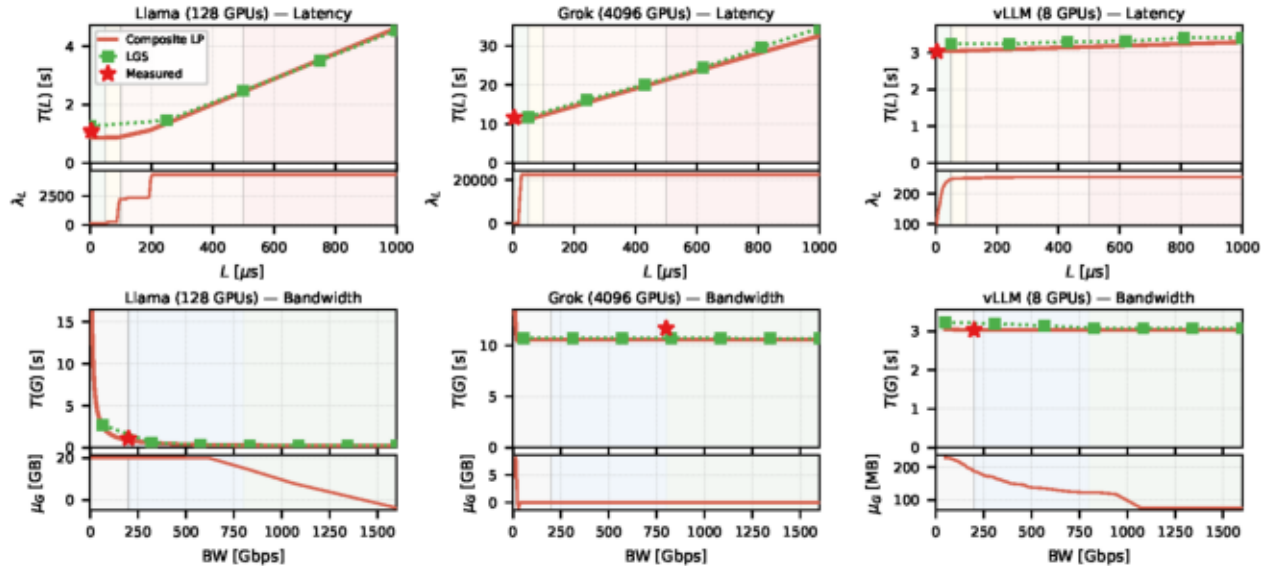
PROVENANCE: the selected Composite curve was produced by an NSYS-derived workflow and is committed in the repository. Building this PDF plots that result without rerunning NSYS. The available raw entry point, pipeline/reproduce_fig5_from_nsys.sh, regenerates the Monolithic baseline rather than this committed Composite curve. Maximum relative error: 0.014853%; mean: 0.004879% over 201 points.

Figure 6: latency and bandwidth sensitivity

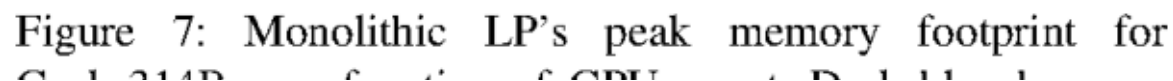
Paper plot



Artifact output: PLOT-ONLY from committed inputs



Paper plot



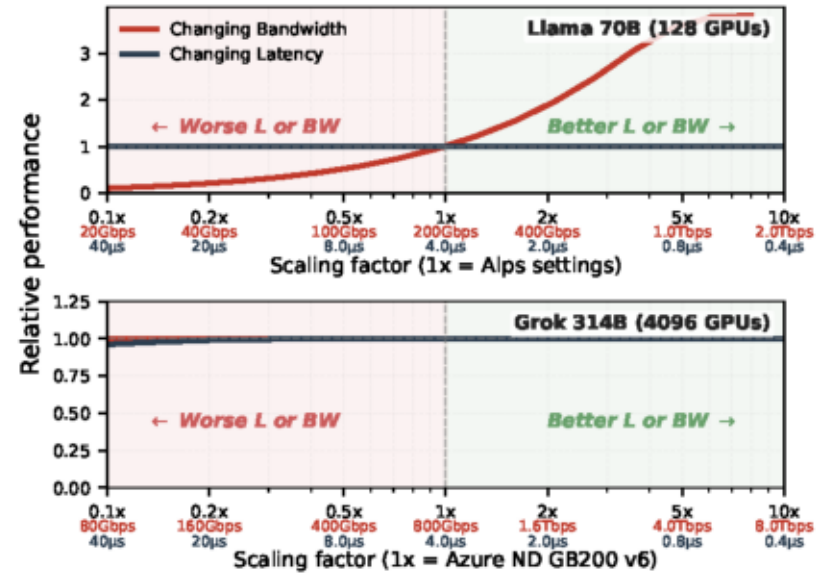
The chart displays Peak Memory [GB] on a logarithmic scale (from 10^1 to 10^5) against the Number of Nodes (N4, N8, N16, N32, N64, N128, N256, N512, N1024). The legend indicates that dark blue bars represent 'Measured' data and light blue bars represent 'Extrapolated' data. A red dashed line with circular markers shows the trend for Monolithic LP, while the bars represent Composition LP. A callout box highlights the memory difference at N1024.

Number of Nodes	Nodes (decimal)	Monolithic LP Peak Memory [GB]	Composition LP Peak Memory [GB]
N4	(16)	8 GB	8 GB
N8	(32)	10 GB	10 GB
N16	(64)	22 GB	22 GB
N32	(128)	59 GB	59 GB
N64	(256)	146 GB	146 GB
N128	(512)	389 GB	389 GB
N256	(1024)	1.7 TB	1.7 TB
N512	(2048)	7.6 TB	7.6 TB
N1024	(4096)	34.0 TB	~0.2 TB

At N1024: Monolithic LP needs ~34 TB vs Composition LP ~0.2 TB (170 times less memory)

Figure 8: network-parameter sensitivity

Paper plot



Artifact output: PLOT-ONLY from committed inputs

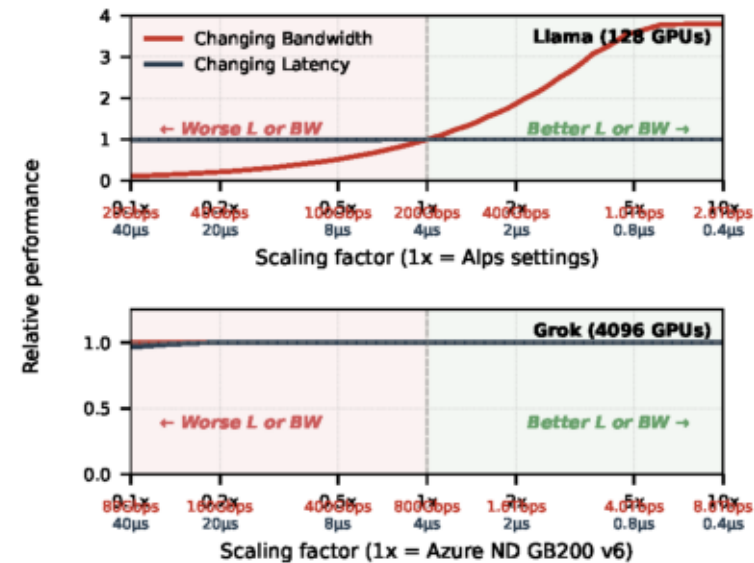
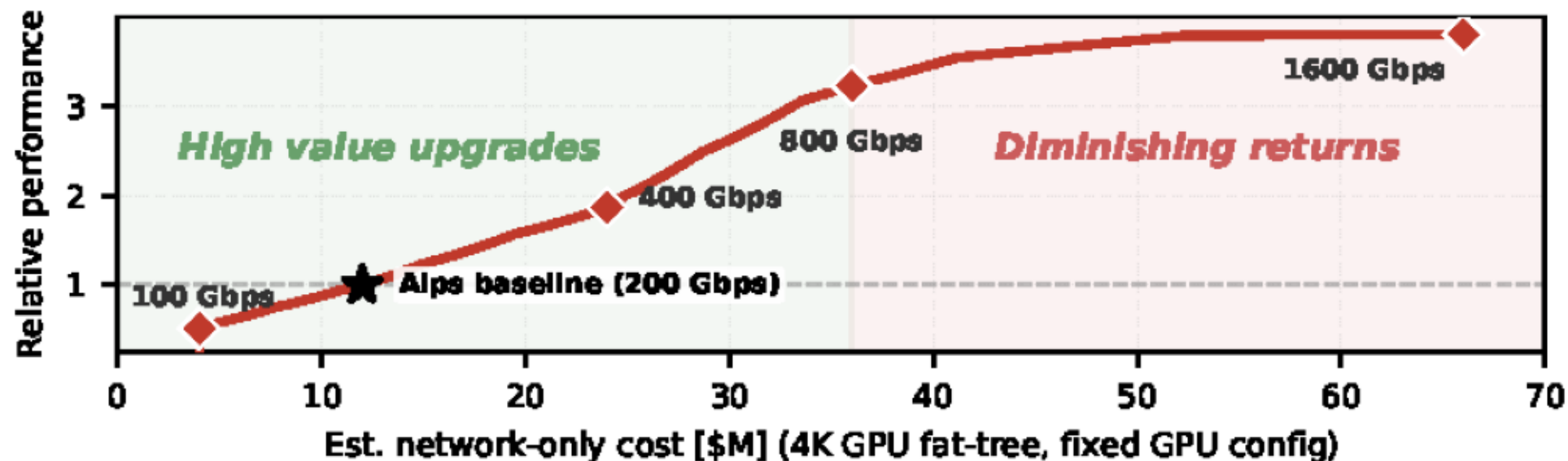


Figure 9: network cost trajectory

Paper plot



Artifact output: PLOT-ONLY from committed inputs

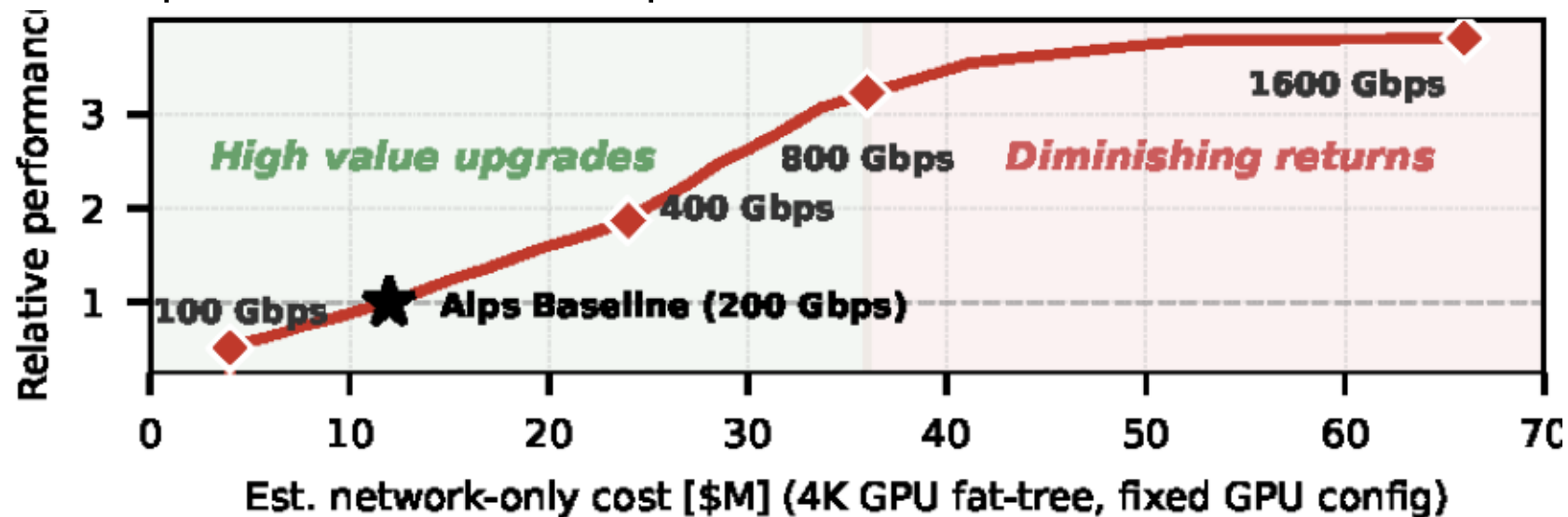
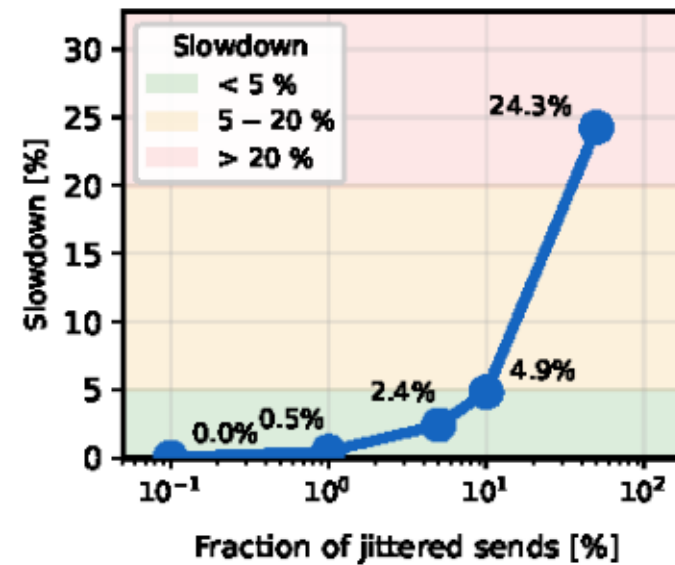
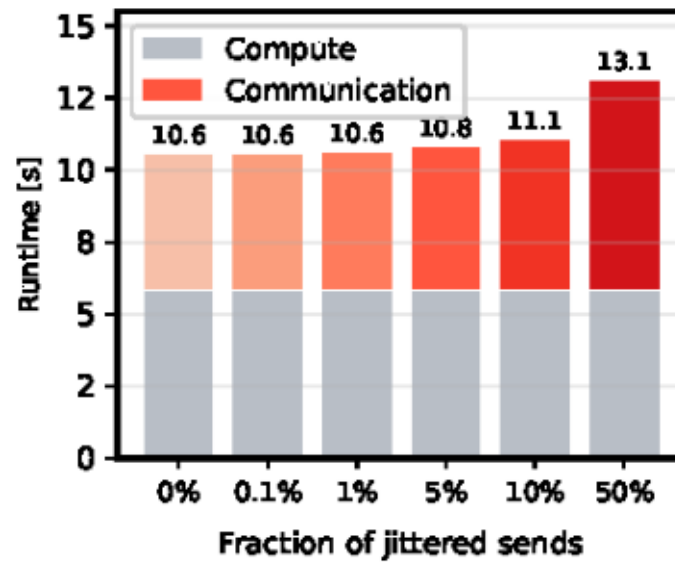


Figure 10: jitter sensitivity

Paper plot



Artifact output: PLOT-ONLY from committed inputs

